

GROUP AND RING THEORY - EXERCISE 13 (ANSWERS)

1. Compute the invariants and write down the structures of the abelian groups with generators $x_1, \dots, x_n$ subject to the following relations:

(a) $n = 2, x_1 + x_2 = 0$;

(b) $n = 3, 3x_1 - 2x_2 = 0, x_1 + x_3 = 0$, and $-x_1 + 3x_2 + 2x_3 = 0$;

(c) $n = 3, 2x_2 - x_3 = 0, -3x_1 + 8x_2 + 3x_3 = 0$, and $2x_1 - 4x_2 - x_3 = 0$.

Solution. (a) We have

$$\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

which corresponds to $\mathbb{Z}$.

(b) Here we have

$$\begin{pmatrix} 3 & -2 & 0 \\ 1 & 0 & 1 \\ -1 & 3 & 2 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix},$$

giving C_3 .

(c) Here

$$\begin{pmatrix} 0 & 2 & -1 \\ -3 & 8 & 3 \\ 2 & -4 & -1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 10 \end{pmatrix} \quad (R_3 - 5R_2, -C_3)$$

giving the invariants 1, 1, 10 and hence the group is $C_{10} \cong C_2 \times C_5$.